Vaibhav Barsaiyan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.80
Central Academy English Medium School , Jbp (MP)	Class XII (CBSE)	2011	69.40%
Central Academy English Medium School , Jbp (MP)	Class X (CBSE)	2009	84.20%

WORK EXPERIENCE

- **Research Assistant (RA)**, @PDCRL, Dept. of CSE, **IIT Hyderabad** (DRDO-AI/ML project)

-Engineered a modular Python/C++ Self-Adaptive streaming-clustering engine for Radar PDW deinterleaving

-Currently integrating it with FPGA techniques for further improvements.

PROJECTS

- **Concurrency Control in Transactional Systems (MVCC-SV & SSI)**

[github.com/Vb-777/Concurrency-Control-in-Transaction-Systems]

- Implemented MVCC-SV and SSI serializable protocols in C++ with an automated benchmark harness

- Benchmarked 36 runs (up to 2,000 txns): SSI capped peak latency at **1.97 s vs 4.49 s** for MVCC-SV

Tech: C++, multithreading, Python (Matplotlib)

- **Parallelised Sequential k-means++ Clustering**

[github.com/Vb-777/Parallelized-Sequential-Kmeans-Clustering-Algorithm]

- Parallelised k-means++ in C++ with Pthreads (partitioned assignment, mutex-merged partial sums)

- Cut runtime from **31.9 s to 6.1 s (5.2×)** on 500K points with 32 threads; **~7×** at k = 100

Tech: C/C++, Pthreads, Python (Matplotlib), Hybrid Validation Metrics, MPI(openMP/pThreads)

- **Adaptive Distributed K-means Clustering (C & DC)**

[github.com/Vb-777/Distributed-KMeans-Clustering]

- Built centralised and decentralised distributed k-means in C++/MPI with automatic k selection

- At 50K points, cut runtime from **790 ms** (sequential) to **408 ms (1.9×)** and **298 ms (2.6×)**

Tech: C++, MPI, Python, JSON, Matplotlib

- **AI/ML-based PDW Emitter Deinterleaving & PR using HPC on FPGA**

[github.com/Vb-777/AI-ML-based-PDW-radar-Emitter-Deinterleaving-and-PR-using-HPC-on-FPGA]

- Built a 9-stage ESA-Stream (IEEE TKDE '20) grid-clustering pipeline for online PDW deinterleaving

- On 32,768-PDW datasets, ran **11-29% faster** than k-means without needing a prior cluster count

- Designed a Zynq UltraScale+ PS-PL partition (stages 1-4 in PL); Vitis HLS port in progress

Tech: Python, MATLAB, C++ (STL), STAPL, Vitis HLS, Vivado, PYNQ

- **SarDet: Diffusion-based SAR Aircraft Detection**

[github.com/Vb-777/SarDet]

- Reproduced DiffusionDet (ResNet-50 + FPN) on SAR-AIRcraft-1.0 at **87.1 mAP@50** (paper: 88.5)

- Improved with 3 novelties: Adaptive SFE, clutter filtering, and contrast-aware thresholding: mAP@50 **87.1** to **89.7**

Tech: Python, PyTorch, Detectron2, OpenCV, Matplotlib, ResNet-50, FPN / SFE / PDC (P5)

- **Contrastive Clustering-based Self-Supervised Adaptation**

[github.com/Vb-777/Contrastive-Clustering-based-Self-Supervised-Adaptation]

- Reproduced PFC source-free domain adaptation (ResNet-50, Office-31 A to W) at **82.64%** accuracy

- Built a contrastive self-supervised variant (projection head, k-means prototypes): **83.31**%

Tech: PyTorch, Torchvision, scikit-learn, ResNet-50, Google Colab (Tesla-GPU)

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**, Dept. of CSE, IIT Hyderabad - CCTS , CS-5280 (Fall-26)

- **Teaching Assistant**, Dept. of CSE, IIT Hyderabad - PCP, CS-5300 (Spring-26)

- **Teaching Assistant**, Dept. of CSE, IIT Hyderabad - Adv. OS, CS-3523 (Spring-25)

- **Mentor SRFP-26 IITH Interns** - PDCR Lab-317 (High Perf. Comp.), IITH CSE

- **IR Cell Coordinator** - Guiding SEP students from Kazakhstan, Nepal, and Japan

- **Edu. Coordinator,** @ **Aksharadaan IITH-SCERT** - Maths sbj in Schools (30+ students)

- Volunteering Student**, GOOGLE** Developer Students Club (GDSC, IIT Ropar)

- Participating Member**, Suzuki Innovation Center-Japan Hour Events** @TRP IITH

- Cultural Events **Volunteer**, **SIC Next Bharat** @TRP IITH

- Team Manager**, Valorant** FPSC 2024 **GLITCH E-sports** Club - IITH **Milan Tech** Festival

- Volunteering Coordinator @**JIGYASA, Vedic Science Club**, IITH

- **Volunteering Participant** - IITH InnerVenture Yoga Meditation Club

- **Volunteering Participant** - IITH Running Enthusiasts Club

- **Volunteering Participant** - IITH Cycling Fitness Club

- Volunteering Participant**, Blood Donation Camp (**BDC**)** - **NSS** IITH

SKILLS

- **Programming Languages** : C++ (STL, OOP) | C | Python

- **Core CS** : Data Structures & Algorithms | TOC | OS | DBMS | Concurrency & Multithreading | Computer Networks

- **Tools & Platforms** : Linux | Git/GitHub | Docker | Apache | Spark | Flink | Shell / Batch scripting

- **Databases**: MySQL | MongoDB

- **Competitive Programming** : 200+ problems solved on LeetCode | GeeksforGeeks | NeetCode

- **Web Tech** : HTML | CSS | PHP

- **Security & Others** : Wireshark | Penetration testing | Network & Web Security

- **Networking** : Switching (VLAN) | Routing (RIP, EIGRP, OSPF) | ACL | NAT / PAT / DAT | DHCP

- **Operating Systems** : Linux (RHEL, Debian, CentOS, Kali, BackTrack R3) | Windows Server 2012 / 2008

CERTIFICATIONS

- **C Programming** Language - Sharma Computer Academy, Bhopal | ID : SCA/Xv/8487

- **C++ Programming** Language - Sharma Computer Academy, Bhopal | ID : SCA/Xv/8486

- **JAVA** Desktop Technology - Vedisoft SES Pvt. Ltd., Bhopal (grades 8/10)

- **Core Java (JSE)** - Sharma Computer Academy, Bhopal | ID : SCA/Xv/993

- **Android** Training (4 weeks) - A²I Pvt. Ltd., Bhopal | ID : A²I/BPL/16/T1754

- **Deep Learning** with **PyTorch**: Image Segmentation - Coursera | GNW25KLB6D27

- 3-Day **AI Accelerate Boot Camp** - Bharat Space Education Research Centre (BSERC)

- Cloud Computing **Major** Training - **Red Hat** Authorised Training Centre (ITE), Bhopal

- CCNA **Cisco** Training (6 weeks) - APPIN Tech Lab, Bhopal | ID: ccna-2016-142

- **Red Hat** RHEL-Linux 5.4 Training (6 weeks) - APPIN Tech Lab, Bhopal | ID: linux-2016-138

- **Networking** Training (4 weeks) - APPIN Tech Lab, Bhopal | ID: N-2016-144

- **MCSE** Training (4 weeks) - APPIN Tech Lab, Bhopal | ID: mcse-2016-149

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

- Top **10** Participant - SIC **Japan** Hour cultural & sports events

- **Bronze Medalist** - IITH Cycling Marathon **10** km (Rank-**3**) **| Finisher** - 21 km

- **Run Marathon**- 5 km IITH 'Nov-25 (top **20**) **|** 5 km IITH 'Nov-26 **|** 3 km iTimes 'Oct-26

- **Cricket** - **Silver** (Feb 2025) | **Gold** (Feb 2026), IITH Inter-Dept. Championship

- **Valorant (GLITCH E-sports)** - **Gold** (IIT Ropar 2023) **| Silver** (IITH 2024) **| Gold** (IITH 2025)

- Top **5** Performer - **FIT** India Yoga Day, IITH Yoga Club

RELEVANT COURSES

Advanced Data Structures & Algorithms | Advanced Operating Systems | Concurrency Control in Transactional Systems | Distributed Computing | Parallel and Concurrent Programming | Theory of Computation | Discrete Mathematics | **Machine Learning** | Probability and Statistics | Linear Optimisation | Data Mining | **Artificial Intelligence** |Fraud Analytics | Visual Computing | Compiler Technologies | **Multimedia Content Analysis** | Network Security | Bitcoin Cryptocurrency |

Internet of Things (IOT) | Topics in Internet Technology | PGSL | Agriculture Cyber-Physical Systems | Computer Architecture | Mathematical FCS | Computer Networks and Systems | Research Methodology in CS | Graph Theory | **Data Analytics Tools & Techniques** | Big Data Tools & Techniques | **Japanese Language**

LANGUAGE LEARNING

- **Languages Known :** English (Fluent) | Hindi (Native)

- **Japanese Language Learner :**

- Spoken Japanese Language **Credited** Course @IITH

- Japanese Learning Student @**SIC** TRP IITH

- Japanese Learning Student @**JECI**, Telangana Center

- Japanese (R/W/L/S) - **JLPT [** N5 **]** Aspirant

INTERESTS

- Research (AiML/Hpc) | Cycling | Running | Badminton | Reading & Writing | Meditation | Yoga | Cricket | Hackathons